

Section B

- 6 (short) A four variable Boolean function is defined as

$$F = AD + B\bar{C}\bar{D} + \bar{A}\bar{B}D + A\bar{B}\bar{C}D.$$

Using a Karnaugh map, or otherwise:

- (a) Find the simplest Sum of Products expression for F . [4]
- (b) Identify under what circumstances a change to a single input variable may cause a momentary unexpected change in F . Give the technical term for this problem. [3]
- (c) Modify the Sum of Products expression found in part (a) to remove this possible problem. [3]